In [3]:
```
import pandas as pd

data = pd.DataFrame({
    'Date': pd.date_range(start='2025-01-01', periods=60, freq='D'),
    'Page_Views': [
        120, 135, 128, 140, 150, 145, 160, 155, 170, 165,
        180, 175, 190, 185, 200, 195, 210, 205, 220, 215,
        230, 225, 240, 235, 250, 245, 260, 255, 270, 265,
        280, 275, 290, 285, 300, 295, 310, 305, 320, 315,
        330, 325, 340, 335, 350, 345, 360, 355, 370, 365,
        380, 375, 390, 385, 400, 395, 410, 405, 420, 415
    ]
})

data.set_index('Date', inplace=True)

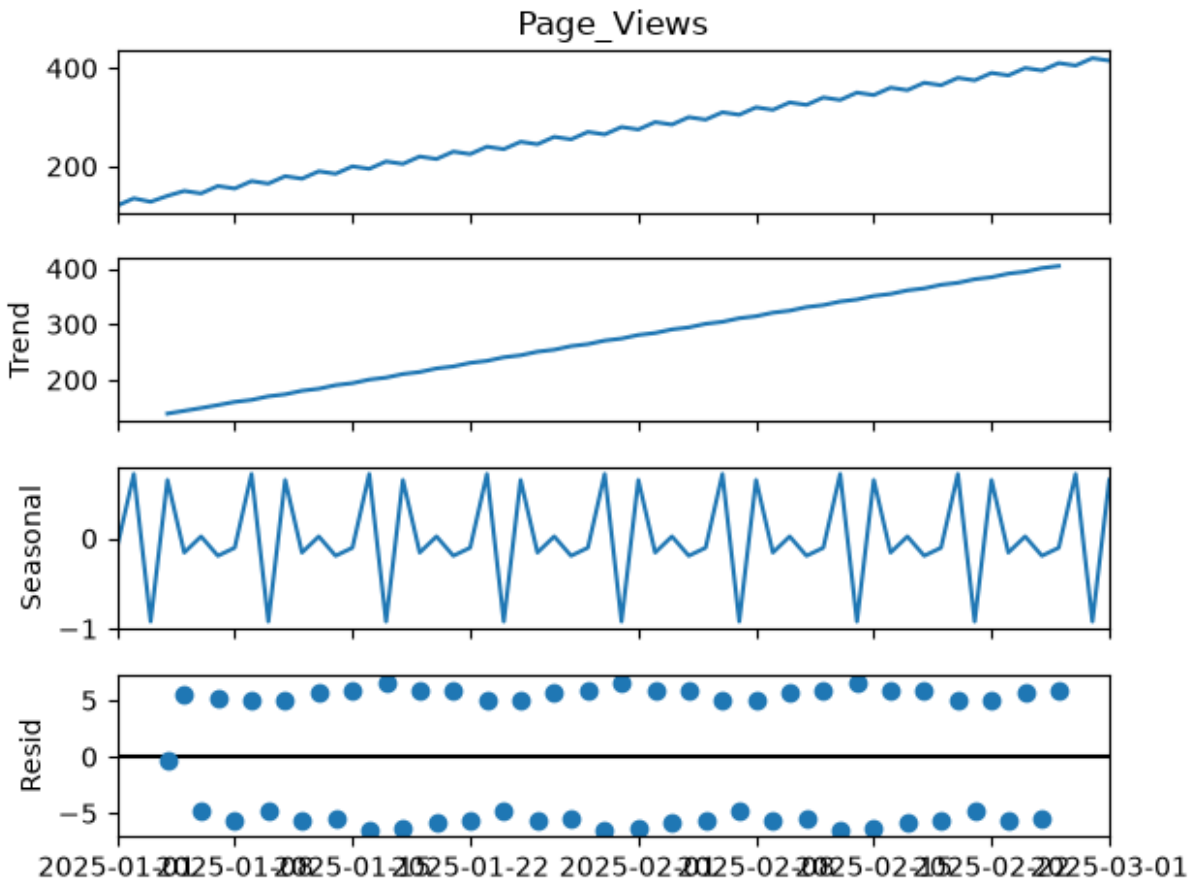
print(data.head())
```

```
            Page_Views
Date
2025-01-01         120
2025-01-02         135
2025-01-03         128
2025-01-04         140
2025-01-05         150
```

In [6]:
```
import matplotlib.pyplot as plt
from statsmodels.tsa.seasonal import seasonal_decompose

decomposition = seasonal_decompose(
    data['Page_Views'],
    model='additive',
    period=7
)

decomposition.plot()
plt.show()
```

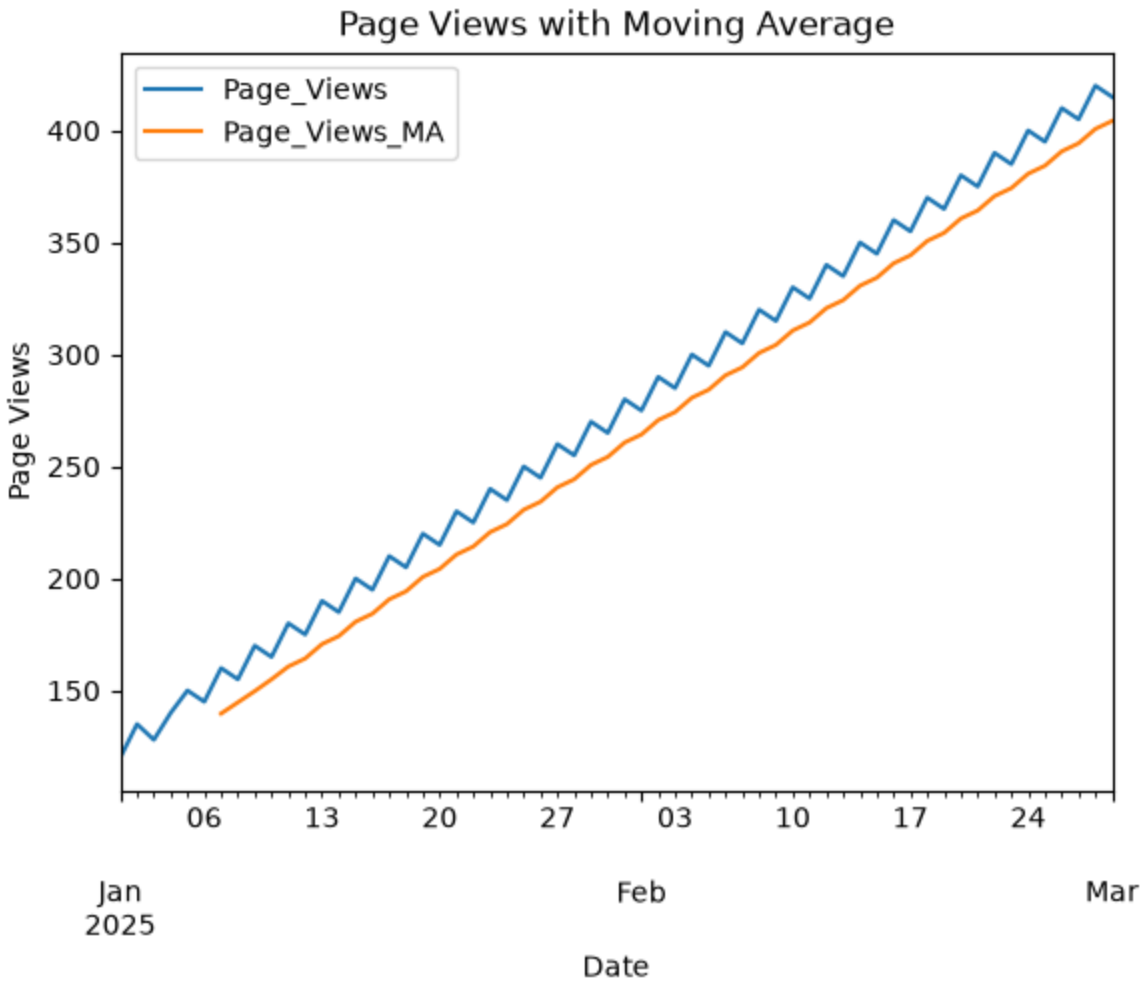


In [7]:
```
import matplotlib.pyplot as plt

data['Page_Views_MA'] = data['Page_Views'].rolling(window=7).mean()

data[['Page_Views', 'Page_Views_MA']].plot()

plt.title('Page Views with Moving Average')
plt.xlabel('Date')
plt.ylabel('Page Views')
plt.show()
```



In [8]:
```
import matplotlib.pyplot as plt

data['Page_Views_Z'] = (
    (data['Page_Views'] - data['Page_Views'].mean())
    / data['Page_Views'].std()
)

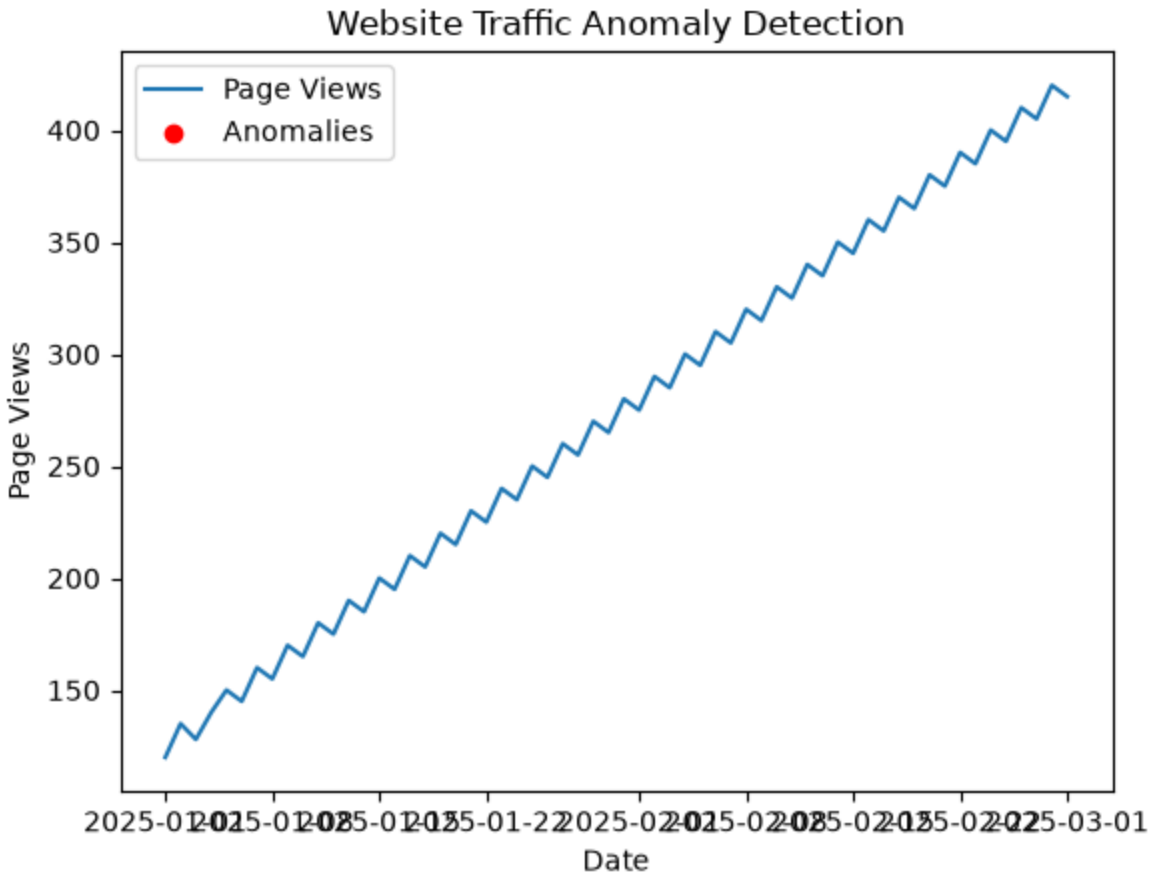
anomalies = data[data['Page_Views_Z'].abs() > 3]

plt.plot(data.index, data['Page_Views'], label='Page Views')

plt.scatter(
    anomalies.index,
    anomalies['Page_Views'],
    color='red',
    label='Anomalies'
)

plt.title('Website Traffic Anomaly Detection')
plt.xlabel('Date')
plt.ylabel('Page Views')
plt.legend()
plt.show()

print(anomalies)
```




```
Empty DataFrame
Columns: [Page_Views, Page_Views_MA, Page_Views_Z]
Index: []
```

In [ ]: